

A δ^2 – L^3 Sensitivity Law for Finite-Window Weil Positivity, with a Function-Field Control of the Connes–van Suijlekom Pipeline

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Abstract

The Connes–van Suijlekom and Connes–Consani–Moscovici programs reduce the Riemann hypothesis to spectral properties of a quadratic form restricted to test functions supported in a finite window. We ask how sensitive that form is to a zero moved off the critical line, and we calibrate the answer in a setting where the truth is known: curves over finite fields, where the window form is exactly the Frobenius Gram matrix of Hallouin–Perret and the Riemann hypothesis is Weil’s theorem. Our main result is a sensitivity law. In the function field we prove that moving a reciprocal real pair $\{\rho, \rho^{-1}\}$ off the critical circle gives $\lambda_{\min}(T_R) = -\binom{R+2}{3}a^2 + O(a^4)$ with $a = \log \rho$, the coefficient being twice the centred second moment of the window. On the zeta side, in the even sector, we prove an unconditional bound $\lambda_{\min}(Q_\delta) \geq -c(L)\delta^2 - O(\delta^4)$ with $c(L) = 8L^3(\frac{1}{3} + \frac{1}{\sqrt{5}})$, valid on windows where positivity is certified, and a sharper conditional bound with the *exact* constant $4K(L, \gamma)$, where $K(L, \gamma) = \int_{-L}^L u^2 \sin^2(\gamma u) du$ is attained by $f \propto u \sin(\gamma u)$; numerically the re-optimised minimiser reaches up to 96% of it. Under an explicit dictionary the two laws are the same expression, (window length)³/6. We also run the full Connes–van Suijlekom minimiser pipeline on deliberately planted off-circle spectra, which separates the steps that carry arithmetic content from those that do not. **Nothing here proves or advances the Riemann hypothesis for ζ .**

Non-claims

Stated first, not in an appendix.

1. **No proof or advance of RH for ζ .** Part I runs where the Riemann hypothesis is a theorem (Weil, curves over finite fields); Part II uses fabricated spectra.
2. **No new spectral architecture.** The window form, its Toeplitz structure and its Carathéodory–Fejér theory are due to Hallouin–Perret [8] and Connes–van Suijlekom [5]; the finite-window Weil form itself goes back to Yoshida [10] and Bombieri [1].
3. **Planted spectra are fabricated diagnostics**, not L -functions of anything.
4. **Numerical detections are tier T2 unless certified**, and *non-detections certify nothing*: a finite basis that finds no negative direction is not evidence of positivity. This asymmetry is the subject of §6.

5. $\lambda^*(0.8) \leq 2.2702 \times 10^{-17}$ **is a variational upper bound** from a finite basis, not a certified enclosure. We prove no rigorous lower bound *of our own*; Zhu [11] certifies one, and Theorems 7–8 rely on it.
6. **§5 reproduces published computations** (Groskin [6]) at smaller scale.
7. **Block F is claimed only for the function field.** On ζ windows the indefiniteness of A and M depends on where $-\log \pi$ is placed; see §7 and Retraction 15.
8. **Not attempted:** Davenport–Heilbronn; any Lean formalisation.

Epistemic tiers

Tier	Meaning	Discharged by
T1	exact or proved	a proof, or exact arithmetic over $\mathbb{Z}$ or $\mathbb{Q}$
T2	numerical	stated working precision + a convergence or noise-floor check
T3	interpretation	argument from T1/T2 evidence, labelled as such

Every numbered statement below carries a tier, the script that produced it and the data file that records it. All numerics use `mpmath` at the stated precision; `float64` is used only as a deliberate contrast (§10).

1 The pipeline and the question

Let $X/\mathbb{F}_q$ be a smooth projective absolutely irreducible curve of genus g , and let $N_n = \#X(\mathbb{F}_{q^n})$. Hallouin and Perret [8, Prop. 5] show that the normalised classes γ^k of the iterated Frobenius graphs have Gram matrix

$$\text{Gram}(\gamma^0, \dots, \gamma^R) = (t(|i - j|))_{0 \leq i, j \leq R}, \quad t(0) = 2g, \quad t(n) = \frac{(q^n + 1) - N_n}{q^{n/2}}, \quad (1)$$

a symmetric Toeplitz matrix, which we write T_R . We adopt (1) as the definition of the window form; it is *their* matrix, entry for entry.

The Connes–van Suijlekom pipeline [5] passes through four steps: (i) reality of the zeros of the ground-state polynomial; (ii) the sign of $\lambda_{\min}$; (iii) simplicity and evenness of the ground state; (iv) convergence of the normalised minimiser. The question of this note is *quantitative*: how large a negative eigenvalue does a zero moved a distance δ off the critical line actually produce, and how large a window is needed to see it?

What is claimed as new. Only two things.

(N1) The δ^2 – L^3 sensitivity law: Theorem 6 (function field), Theorems 7 and 8 (ζ), the measured near-saturation, and the shared (window length)³/6 form.

(N2) The planted-spectrum control of the CvS minimiser pipeline in the function field (§3).

Everything else is foundation or confirmation; §8 says exactly whose.

2 Foundation (prior art, stated as such)

Proposition 1 (Hallouin–Perret [8, Prop. 5, §1.2]). *[T1, cited] The Gram matrix of the normalised iterated-Frobenius classes is the Toeplitz matrix (1).*

Theorem 2 (Hallouin–Perret [8, Thm. 6, §1.3]). *[T1, cited] Let d be the rank of the Frobenius space F . Then d is the minimal integer such that $\text{Gram}(\gamma^0, \dots, \gamma^d)$ is singular, a generator of that kernel exists over $\mathbb{Q}$, and the resulting Frobenius eigenvalues are algebraic integers of modulus $\sqrt{q}$.*

Theorem 3 (Hallouin–Perret [8, Thm. 36(ii), App. A.2]). *[T1, cited] If a real symmetric positive semi-definite Toeplitz matrix of size $n + 1$ has rank n and kernel vector $(a_0, \dots, a_n)$, then $P = \sum a_j X^j$ has n distinct roots, all of modulus 1.*

Theorem 3 is step (i) of the pipeline; it is the Carathéodory–Fejér corollary that CvS [5, Cor. 1.1] prove again C^* -algebraically. Their Lemma 33 (rank = largest non-zero *leading* minor) and Lemma 34 (the $(\gamma_i \pm \gamma_{n-i})/\sqrt{2}$ parity split) are the computational tools we use throughout. CvS themselves point to [8] for the function-field connection and write that “the key difficulty in this context, then, becomes the verification that zero is indeed the (simple) minimal eigenvalue of T ” [5, Introduction].

3 The control experiment

3.1 Genuine spectra

[T1 for the arithmetic, T2 for the spectra; `src/stage1.py`, `data/stage1_spectra.csv`] Four curves ($y^2 = x^3 + x/\mathbb{F}_5$, $y^2 = x^5 + x^2 + 1/\mathbb{F}_3$, $y^2 = x^7 + x + 1/\mathbb{F}_5$, $y^2 = x^7 + x^3 + x + 1/\mathbb{F}_3$; $g = 1, 2, 3$) with brute-force point counts, L -polynomials by Newton’s identities plus the functional equation, and exact integer power sums. All 40 rows have $\lambda_{\min} \geq 0$; $\lambda_{\min}(R)$ decreases monotonically (Cauchy interlacing) and vanishes exactly at $R = d$, where the kernel polynomial’s roots are the $e^{i\theta_j}$ — Theorem 2 in action.

3.2 Planted spectra

[T1, exact over $\mathbb{Q}$; `src/stage2.py`, `src/exact_inertia.py`] We replace the Frobenius spectrum by a functional-equation-closed multiset with $|\beta| \neq 1$. Choosing rational ρ and $\cos \varphi$ makes every $t(n)$ rational, so the inertia is certified *exactly* by the Sylvester–Jacobi leading-minor rule.

configuration	planted	first R with $\lambda_{\min} < 0$	step (i) holds?	$\lambda_{\min}$ simple?
on-line control	$ \beta = 1$	never (PSD, proved)	always	fails for every $R > d$
real pair	$\rho = 1.3$	$R = 1$	always	never fails
quartet	$\rho = 1.3, \varphi = 0.7$	$R = 2$	always	never fails
two quartets	$\rho = 1.3$	$R = 4$	always	never fails
mixed	quartet + pair	$R = 4$	always	never fails

Observation 4. [T1 reasoning, T2 evidence] Step (i) holds at every applicable row of every planted configuration, including rows where $\lambda_{\min} = -642.8$: measured $\max||z| - 1| \leq 5.1 \times 10^{-48}$. Steps (i) and (iii) are therefore free of arithmetic content. This is not a discovery — CvS state Theorem 3 for an arbitrary real even distribution, so no arithmetic can enter step (i) — but the planted control makes it concrete, with numbers attached.

3.3 Degeneracy beyond the window degree

[T1 corollary of Thm. 3(i); `src/function_level.py`] Since the shift has P as both minimal and characteristic polynomial, $\ker T_R$ is the ideal generated by P truncated at degree R :

$$\ker T_R = \{P \cdot q : \deg q \leq R - d\}, \quad \dim \ker T_R = \max(0, R + 1 - d), \quad (2)$$

matched in every computed row. Hence $\lambda_{\min} = 0$ is simple exactly at $R = d$ and never after, so the CvS simple-ground-state hypothesis fails for every $R > d$; and since P is self-reciprocal, the kernel splits into even and odd parts of dimensions $\lceil \frac{m+1}{2} \rceil, \lfloor \frac{m+1}{2} \rfloor$ with $m = R - d$, both non-zero for $m \geq 1$. CvS state this in words in their Remark 2.3; we record the dimensions.

3.4 Step (iv) at the function level

[T1; src/function_level.py]

Proposition 5. *If $\lambda_{\min}$ is simple for infinitely many R and the normalised ground-state polynomials converge locally uniformly to $F \not\equiv 0$, then every zero of F lies on $|z| = 1$ by Theorem 3 and Hurwitz’s theorem. Hence they cannot converge to a target with a zero off the circle.*

The free witness is exactly the obstruction. Measured: zero *angles* converge to $\sim 10^{-3}$ even for off-line spectra, while the coefficient distance to the target is exactly $\sqrt{2}$ (orthogonal) and the normalised $|\hat{c}_R(\beta)|$ *risers* to 0.58–0.71 out to $R = 36$ instead of falling. This makes concrete the Hurwitz step that is CvS’s own Step 5.

4 The δ^2 – L^3 law

4.1 Function field

Theorem 6. [T1; proof in notes/proofs.md, checked by src/proofs_check.py] *Let the normalised spectrum be a single off-circle real pair $\{\rho, \rho^{-1}\}$ and put $a = \log \rho$. With $u_i = \rho^i$, $w_i = \rho^{-i}$ ($0 \leq i \leq R$),*

$$T_R = uw^\top + wu^\top, \quad \lambda_{\min}(T_R) = (R+1) - \|u\| \|w\|,$$

all other eigenvalues being 0. Consequently $\lambda_{\min} < 0$ for every $R \geq 1$ as soon as $\rho \neq 1$, by strict Cauchy–Schwarz; $\lambda_{\min}$ is an even function of a ; and

$$\lambda_{\min}(T_R) = -2a^2 \sum_{i=0}^R \left(i - \frac{R}{2}\right)^2 + O(a^4) = -\binom{R+2}{3} a^2 + O(a^4).$$

For $R = 1$ this is exact: $\lambda_{\min} = -4 \sinh^2(a/2)$.

The coefficient is twice the *centred* second moment of the window; the centring falls out of the optimisation, not from a convention. Checks: the factorisation to 6.2×10^{-61} , the eigenvalues to 1.0×10^{-59} , the moment identity exactly for $R = 1, \dots, 199$, and $-\lambda_{\min}/a^2 = \binom{R+2}{3}$ to twelve printed digits for $R = 1, 2, 4, 8, 16, 32$.

4.2 Zeta side

Throughout this subsection f is real, **even**, supported in $[-L, L]$ with $\|f\|_2 = 1$, and $F = \hat{f}$; all statements here are for the even sector.

The baseline, stated explicitly. An off-line zero forces a quartet — four zeros — whereas a *simple* on-line zero at γ contributes a pair. The zeros of ζ are expected to be simple, and every computed one is, so a count-preserving comparison must be made against a *hypothetical double* on-line zero. Writing Q_{true} for the actual Weil form, the baseline below is

$$Q_0 = Q_{\text{true}} + 2F(\gamma)^2. \tag{3}$$

Two consequences, both used. First $Q_0 \geq Q_{\text{true}}$, so Theorem 7 is unaffected: a lower bound certified for Q_{true} is also one for Q_0 . Second, Theorem 8’s hypothesis $\tilde{Q} := Q_0 - 4F(\gamma)^2 \geq 0$

reads $Q_{\text{true}} - 2F(\gamma)^2 \geq 0$, i.e. exactly “every zero *other* than the one being moved lies on the critical line”. Of the three conventions this is the only one with $Q_\delta \rightarrow Q_0$ as $\delta \rightarrow 0$, and (3) is what that continuity costs: it is bought by shifting the baseline, not by comparing against ζ itself.

Moving the double zero to $\frac{1}{2} + \delta \pm i\gamma$ changes the form by

$$Q_\delta(f) - Q_0(f) = -4\delta^2(F'(\gamma)^2 + F(\gamma)F''(\gamma)) + R_4. \quad (4)$$

Theorem 7 (unconditional on certified windows; even sector). [T1] With $c(L) = 8L^3(\frac{1}{3} + \frac{1}{\sqrt{5}}) \approx 6.244L^3$,

$$\lambda_{\min}(Q_\delta) \geq \lambda^*(L) - c(L)\delta^2 - \frac{16}{3}L^5\delta^4e^{2L\delta}.$$

This is unconditional exactly where Weil positivity on the window has been certified: for $L \leq 0.8$, since Zhu [11] certifies $\lambda^*(0.8) \geq 8.9 \times 10^{-18}$ and λ^* is non-increasing in L . For $L > 0.8$ it holds with $\lambda^*(L)$ replaced by 0, using no positivity input at all.

Theorem 8 (sharper, conditional; even sector). [T1] Suppose every nontrivial zero other than the one being moved lies on the critical line, so that $\tilde{Q} := Q_0 - 4F(\gamma)^2 \geq 0$. Completing the square, $4F^2 - 4\delta^2FF'' = (2F - \delta^2F'')^2 - \delta^4F''^2$, moves the FF'' allowance into $O(\delta^4)$ and gives

$$\lambda_{\min}(Q_\delta) \geq -4K(L, \gamma)\delta^2 - \left(\frac{2}{5} + \frac{16}{3}e^{2L\delta}\right)L^5\delta^4,$$

where $K(L, \gamma)$ is the exact value of the extremal problem,

$$K(L, \gamma) = \sup_{\substack{f \text{ real, even, } \text{supp} f \subset [-L, L] \\ \|f\|_2=1}} |F'(\gamma)|^2 = \int_{-L}^L u^2 \sin^2(\gamma u) du, \quad (5)$$

attained by $f \propto u \sin(\gamma u)$, with closed form

$$K(L, \gamma) = \frac{L^3}{3} - \left[\frac{L^2 \sin 2\gamma L}{2\gamma} + \frac{L \cos 2\gamma L}{2\gamma^2} - \frac{\sin 2\gamma L}{4\gamma^3} \right].$$

Corollary 9 (γ -uniform form). [T1] Since $\sin^2 \leq 1$ gives $K(L, \gamma) \leq \int_{-L}^L u^2 du = 2L^3/3$ for every γ ,

$$\lambda_{\min}(Q_\delta) \geq -\frac{8L^3}{3}\delta^2 - O(\delta^4),$$

which is the form to quote when γ is not fixed. As $\gamma \rightarrow \infty$, $K(L, \gamma) \rightarrow L^3/3$, so the uniform constant overshoots the large-ordinate value by exactly the factor 2 coming from the average of $\sin^2$.

Remark 10. $4K(L, \gamma)$ is sharp: by (5) it is attained, so no smaller δ^2 coefficient is available at that γ . The constants $8L^3/3$ and $4L^3/3$ appearing below are respectively the γ -uniform upper bound and the $\gamma \rightarrow \infty$ limit of the same quantity, and ratios reported against them should be read as such.

4.3 How close the measured constant comes

[T2, src/task_final_checks.py, data/stage3_exact_constant_K.csv]

The planted ordinate is $\gamma_1 = 14.1347$. Measured $C(L)$ against the exact constant $4K(L, \gamma_1)$ of Theorem 8, and for reference against the two constants of Corollary 9:

L	0.8	1.0	1.3	1.6	2.0
measured $C(L)$	0.3108	0.8530	2.4987	4.9218	10.083
$4K(L, \gamma_1)$	0.7419	1.3427	3.1158	5.1130	10.652
$C/4K(L, \gamma_1)$	0.419	0.635	0.802	0.963	0.947
$C/(8L^3/3)$ (<i>uniform</i>)	0.228	0.320	0.427	0.451	0.473
$C/(4L^3/3)$ ($\gamma \rightarrow \infty$)	0.455	0.640	0.853	0.901	0.945

Against the exact constant the re-optimised minimiser reaches 96.3% at $L = 1.6$.

How much the $\sin^2$ weight matters, measured. The deviation $|K(L, \gamma_1)/(L^3/3) - 1|$ is

$$17.3\%, \quad 8.68\%, \quad 0.699\%, \quad 6.37\%, \quad 6.38\%, \quad 0.136\%$$

at $L = 0.6, 0.8, 1.0, 1.3, 1.6, 2.0$. It is *not* monotone: the bracketed terms in (5) oscillate in L with period π/γ , so the correction can be large at one window and negligible at a slightly larger one. At the window where positivity is certified, $L = 0.8$, it is 8.68%; at $L = 2$ it happens to be 0.136%. This is why the γ -uniform constant should not be used as a stand-in for K at a fixed ordinate.

4.4 One law, two settings: the dictionary

[T3, checked numerically in `src/task_final_checks.py`]

The two laws are the same statement under an explicit change of variables. For a curve over $\mathbb{F}_q$ the explicit-formula variable is $\log$ of a prime power, so the lattice index n and the ζ variable u correspond by $u = n \log q$. This fixes everything else:

	function field	ζ
window variable	$n = 0, \dots, R$	$u \in [-L, L]$
correspondence	$u = n \log q$, hence window length	$R \log q = 2L$
off-line depth	$a = \log \rho$	$\delta = \operatorname{Re} \rho - \frac{1}{2}$, with $a = \delta \log q$
Rayleigh quotient	sum over n	integral $du = \log q \, dn$: differ by $\log q$

Substituting $R = 2L/\log q$ and $a = \delta \log q$ into Theorem 6 and dividing by $\log q$:

$$-\binom{R+2}{3} a^2 \approx -\frac{R^3}{6} a^2 = -\frac{1}{6} \left(\frac{2L}{\log q} \right)^3 (\delta \log q)^2 = -\frac{4L^3}{3} \frac{\delta^2}{\log q},$$

i.e. exactly the ζ law with the large-ordinate constant $4L^3/3$, divided by the $\log q$ of the last table row. Equivalently both read (window length) $^3/6$ in their own units.

Which factors of 2 cancel. Two enter, in opposite directions, and they cancel:

1. *Pair versus quartet.* Theorem 6 moves a reciprocal *pair*; the ζ counterfactual moves a *quartet*, twice as many zeros — a factor 2 up.
2. *$\sin^2$ averaging.* The ζ constant uses $K(L, \gamma) \rightarrow L^3/3$, half of the $\int u^2 = 2L^3/3$ that the function-field centred moment is the discrete analogue of — a factor 2 down.

Checked on one curve. For $y^2 = x^3 + x$ over $\mathbb{F}_q$, $q = 5$, with $a = 10^{-6}$, the ratio of the exact function-field $\lambda_{\min}$ to the ζ -law prediction $-\frac{4}{3}L^3\delta^2/\log q$ is

$$1.406, \quad 1.195, \quad 1.096, \quad 1.047, \quad 1.024 \quad \text{at } R = 8, 16, 32, 64, 128,$$

converging to 1 like $1 + 3/R$ — which is precisely $\binom{R+2}{3}/\frac{R^3}{6} = (1 + \frac{1}{R})(1 + \frac{2}{R})$, the only discrepancy being the $O(R^2)$ term dropped in passing from the binomial to $R^3/6$. The dictionary is therefore exact at leading order, with a fully accounted correction.

5 The ζ side as a control

[T2; src/stage3_converge.py, src/stage3_connes.py]

We assemble the window form from Zhu’s geometric side in three closed-form bases (two even, one odd), moving everything to u -space by Parseval so no oscillatory quadrature is needed, and summing the archimedean m -series analytically via digamma, Lerch Φ and Hurwitz ζ . The odd-sector pole term $-2P_j P_k$ — the sign flips because $F(-i/2) = -F(i/2)$ for odd f — is derived here.

Gate A. Under RH the explicit formula gives $Q(f) = \sum_{\rho} |F(\gamma_{\rho})|^2$. The assembled geometric side matches the zeros side to 1.63×10^{-7} (even, vanishing at $\pm L$), 2.26×10^{-7} (odd) and 4.23×10^{-3} (full cosine, which jumps at $\pm L$) at $K = 800$ zeros, each decaying at the rate that basis’s smoothness predicts (γ_K^{-3} versus γ_K^{-1}). *Control:* flipping the pole sign inflates the discrepancy to 5.35 and 0.212, so the test discriminates.

Gate B. $\lambda_{\min}(N)$ decreases monotonically in two independent complete even bases to 2.2702×10^{-17} and 2.3357×10^{-17} at $N = 24$, landing on Zhu’s certified $[8.9 \times 10^{-18}, 2.27 \times 10^{-17}]$. Zhu normalises by $Q(f)/\|f\|^2$, the same normalisation used here. We report the inequality, not the matching digits: the sequence is still falling.

Connes’s window. At $2L = \log 13$, $N = 36$, 140 digits: $\lambda_{\min} = 8.977 \times 10^{-52}$, simple and even, with the ground state’s Fourier zeros matching γ_1 to 7.2×10^{-48} and tracking roughly $N/2$ leading zeros. This reproduces at smaller scale what Groskin [6] computes at $N = 100$ –250.

Simplicity and evenness. [T2, $N = 12$, 40 digits] At $L \in \{0.5, 0.8, 1.0, \frac{1}{2} \log 13\}$ the ground state is simple with relative gap 2×10^4 to 4×10^5 and *even* in every case, the odd sector sitting 2–5 orders above. The $L = 0.8$ entry is consistent with Zhu’s certificate; the other three carry no external certificate and are T2 only.

6 Detection onset

[T2; src/stage3e_certify.py, data/stage3e_certified_onset.csv]

Precedent. Bombieri [1, §13] already ran this experiment in 2000: the first $N \leq 160$ zeros of ζ together with “a fictitious zero ρ_0 off the critical line, together with their images by complex conjugation and reflection” — the same quartet — at $\rho_0 = 0.52 + 3.14i$, on windows $E = [-t, t]$, and found a critical value t_c beyond which the negative eigenvalue stays bounded away from 0. He also separates even and odd eigenfunctions. Our contribution is to resolve that critical window in δ and in basis size, and to identify it with Zhu’s bandwidth $T^*(L) = 2\pi e^{2L}$.

The certification asymmetry. A negative Rayleigh quotient from *any* trial function certifies indefiniteness. A finite basis that finds none certifies nothing. We therefore grow the basis (low block $\cup$ resonant block $\{k : w_k \approx \gamma\}$) and report, for each planted ordinate, the smallest window at which *some* trial function goes negative. The geometric side supplies the on-line part exactly; truncating the zeros sum omits positive mass $\sim 10^{-6}$ against a signal $\sim 10^{-27}$ and cannot certify anything near the onset.

planted at	γ_n	$L_{\text{pred}} = \frac{1}{2} \log \frac{\gamma}{2\pi}$	onset, small basis	onset, largest basis
γ_1	14.13	0.405	$1.3\times$	$1.3\times$
γ_5	32.94	0.828	$1.0\times$	$1.0\times$
γ_{10}	49.77	1.035	not found	$1.0\times$
γ_{30}	101.3	1.390	not found	$0.85\times$
γ_{80}	201.3	1.733	not found	$1.15\times$

The onset tracks L_{pred} within about $\pm 30\%$ and moves *earlier* as the basis grows. Rows whose magnitude falls below the measured noise floor (the geometric control is PSD by construction, so a negative value there *is* the floor) are excluded and flagged in the data file.

What detection actually needs — and what it does not. Two conditions are visible in the data: the window must reach the ordinate, $L \gtrsim L_{\text{pred}}(\gamma_n)$, and the basis must *contain modes resonant with γ_n* , i.e. indices near $k^* \approx \gamma_n L / \pi$ — which is what the resonant block supplies. It does *not* require a basis large enough to annihilate the first n zeros. The dimension needed is far below the top mode index: γ_{30} is detected with dimension 46 and top index 45, and γ_{80} with dimension 59 and top index ≈ 134 , whereas annihilating the first n zeros in a contiguous basis would need $N \approx 2n$, i.e. 60 and 160. The $N/2$ rule of §5 governs a different question — how many leading zeros a *contiguous* basis can annihilate — and is not the detection criterion. Detection only needs some f in the subspace with $Q_{\text{geom}}(f) + 2F(\gamma)^2 < 4\delta^2(F'^2 + FF'')$, which the sparse “low block $\cup$ resonant block” subspace achieves at a small fraction of the dimension a contiguous basis would need.

Weak δ -dependence, explained. Detection is $C(L)\delta^2 > \lambda^*(L)$. By Zhu’s Landau–Widom law $\lambda^*(L)$ falls super-exponentially in L while $C(L) \sim L^3$ grows polynomially, so a factor in δ^2 shifts the crossing only logarithmically. Measured: a tenfold change in δ moves the onset from $L = 0.5$ to $L = 0.6$.

7 The recorder split

[T2; src/task2_inertia_reconcile.py]

Write $T_R = A - 2M$. In the function field the pole part of A is $q^{|n|/2} + q^{-|n|/2} = u_i w_j + w_i u_j$, an exact rank-2 *hyperbolic* plane whose scale grows like $q^{R/2}$ against a fixed $2g \cdot I$; so A is eventually indefinite under both conventions tested, M is indefinite, and $S = T_R$ is PSD. This is structural.

On ζ windows it is not. With $A = \text{Pole} + \text{Arch} - \log \pi \cdot I$, $2M = \text{Prime}$ one finds A with negative index 1 for $L \leq 0.8$ and 2 at $L = \frac{1}{2} \log 13$, identical in all three bases. But grouping $-\log \pi$ with the primes instead makes **both A and M positive definite** at $L \leq 0.8$. The reason is structural: for even f the ζ pole term is $h(i/2) + h(-i/2) = 2F(i/2)^2 \geq 0$, rank one and *definite*, not hyperbolic. There is no hyperbolic plane on the ζ side to find. We therefore state the convention ($A = \text{Pole} + \text{Arch} - \log \pi \cdot I$) and claim Block F only for the function field.

8 Relation to prior work

Hallouin–Perret [8] is the function-field foundation: Proposition 5 is the window Toeplitz matrix, Theorem 6 the first-singular-size and kernel generator, Theorem 36(ii) step (i), Lemmas 33–34 the rank rule and parity split used here. **Connes–van Suijlekom** [5] supply the C^* -algebraic proof of step (i) and the Hurwitz step, cite [8] for the function-field connection, and state the simplicity difficulty themselves; their Remark 2.3 is our degeneracy observation in words. **Bombieri** [1] is the precedent for planted ζ experiments (§13) and proves the negative-eigenvalue

count (Theorem 8); **Yoshida** [10] first studied the Weil hermitian form on functions supported in $[-t, t]$. **Zhu** [11] supplies the certified $\lambda^*(0.8)$, the normalisation and the bandwidth T^* . **Groskin** [6, 7] supplies the large-scale Galerkin computations and the archimedean tail order. **Connes–Consani–Moscovici** [3], **Connes** [4], **Connes–Consani** [2] and **Suzuki** [9] are the surrounding program. A fuller account, with verbatim quotations and the verification method, is in `PRIOR_ART.md`.

9 Limitations

1. No rigorous lower bound on $\lambda^*(L)$ of our own; Theorem 7 is unconditional only where Zhu’s certificate reaches ($L \leq 0.8$).
2. Theorem 8 is conditional on all other zeros being on the line.
3. Interval certification covers the zeros-side assembly only: `mpmath`’s interval context has no digamma and no Lerch transcendent, so the geometric side could not be interval-evaluated. What is certified instead is the $\text{LDL}^\top$ witness identity $v^\top Z v = D_j$ and a measured noise floor.
4. Onset non-detections are limited by the subspaces actually searched: the largest had dimension 59 with top mode index ≈ 134 . A non-detection bounds that search, not the method.
5. $C(L)$ is measured at finite δ and finite basis; the saturation $C/(4L^3/3) \rightarrow 1$ is observed, not proved.
6. The ζ -side numbers depend on the planted ordinate, the basis and the truncation.

10 Traps this project documents

The retraction log (Appendix A) is part of the result. The reusable lessons: **(a)** truncating the zeros sum biases the form *downward* and manufactures false detections; **(b)** moving a zero and adding a zero are two different counterfactuals, both legitimate — Bombieri studies the second — and the sharp law needs the first; **(c)** `float64` returns spurious negative eigenvalues of size 10^{-16} where the true value is exactly 0; **(d)** interval libraries have gaps, and `mpmath`’s `iv.sinmpi` is itself broken (it calls a non-existent `ctx.sinmpi`); **(e)** a non-detection certifies nothing; **(f)** keeping known-good regression values was the only thing that caught three sector-rename bugs.

A Retractions

Seventeen corrections were logged during this work; the full list with evidence is in `REPORT.md`. Those a reader must see:

#	Retraction
1	Frobenius radius computed on P rather than its reverse (0.8 instead of $\leq 1.3 \times 10^{-51}$).
3	A tolerance-based sign test misclassified exact zeros; replaced by an error-based test plus exact rational certificates.
4	H2 as first stated was refuted by my own adversarial pass ; restated with a distinctness hypothesis. The restatement is Theorem 2.
6	“Step (iv) is free” retracted — it had been scored on zero angles, which is the wrong object. At the function level it carries content (Proposition 5).

#	Retraction
8–9	A 40-digit run discarded; three sector-rename bugs caught only by regression values.
11	“Low-lying zeros only” retracted as structural. The height sweep had used a truncated on-line side, and non-detections were basis-limited. Redone, $\gamma_{30} \approx 101$ is detected at $0.85 L_{\text{pred}}$. <i>Amended:</i> the added-quartet counterfactual is legitimate in its own right — it is Bombieri’s — and was not the error.
12	γ_{80} “detections” below their own noise floor discarded; later detected at $1.15 L_{\text{pred}}$ with a positive control.
15	“Block F holds on the ζ side” retracted (§7).
16	Three claimed contributions are published theorems: step (i) is [8, Thm. 36(ii)], the first-singular-size result is [8, Thm. 6], the window matrix is [8, Prop. 5]; the degeneracy observation is [5, Rem. 2.3]; the negative-index count is [1, Thm. 8]; the planted experiment is [1, §13]. An earlier abstract-level novelty check had missed all of these.
17	The $N \gtrsim 2n$ detection rule retracted (§6). It contradicted both the stated limitation and the γ_{80} table row. The γ_{80} detection used dimension 59, top index ≈ 134 , against $2n = 160$; γ_{30} used dimension 46, top index 45, against $2n = 60$. The rule had been imported from §5, where $N/2$ governs a stronger question — how many leading zeros a <i>contiguous</i> basis annihilates. The table row and the limitation were correct; only the rule was wrong.

B Reproduction

Code and data: <https://github.com/innerlightr-wq/weil-window-control>. Python 3.12, mpmath 1.3.0, numpy 1.26.4, matplotlib 3.8+; no gmpy2 (installing it speeds the high-precision runs substantially).

Result	Script	Data
§3 genuine	src/stage1.py	stage1_spectra.csv
§3 planted	src/stage2.py	stage2_spectra.csv
§3 kernel	src/function_level.py	stage2_function_level.csv
Thm 6, 7, 8	src/proofs_check.py	stage3_sharp_bound.csv
C vs Cauchy–Schwarz	src/task34_extra.py	stage3_C_vs_CS.csv
$K(L, \gamma)$, dictionary	src/task_final_checks.py	stage3_exact_constant_K.csv
Gates A, B	src/stage3_converge.py	stage3_gateA/B_*.csv
Connes window	src/stage3_connes.py	stage3b_connes_zeros_N36.csv
Detection onset	src/stage3e_certify.py	stage3e_certified_onset.csv
Recorder split	src/task2_inertia_reconcile.py	stage3d_inertia_reconciled.csv

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References

- [1] E. Bombieri, *Remarks on Weil's quadratic functional in the theory of prime numbers, I*, Atti Accad. Naz. Lincei Cl. Sci. Fis. Mat. Natur. Rend. Lincei (9) Mat. Appl. **11** (2000), no. 3, 183–233.
- [2] A. Connes and C. Consani, *Weil positivity and trace formula the archimedean place*, Selecta Math. (N.S.) **27** (2021), no. 4, art. 77. doi:10.1007/s00029-021-00689-4; arXiv:2006.13771.
- [3] A. Connes, C. Consani and H. Moscovici, *Zeta spectral triples*, arXiv:2511.22755.
- [4] A. Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, arXiv:2602.04022.
- [5] A. Connes and W. D. van Suijlekom, *Quadratic forms, real zeros and echoes of the spectral action*, arXiv:2511.23257.
- [6] A. Groskin, *High-precision approximation of Riemann zeros via the truncated Weil form*, arXiv:2605.20224. doi:10.5281/zenodo.19546514.
- [7] A. Groskin, *A finite Guinand–Weil dictionary and archimedean tail order for the truncated Weil quadratic form*, arXiv:2607.02828.
- [8] E. Hallouin and M. Perret, *A unified viewpoint for upper bounds for the number of points of curves over finite fields via Euclidean geometry and semi-definite symmetric Toeplitz matrices*, Trans. Amer. Math. Soc. **372** (2019), no. 8, 5409–5451. doi:10.1090/tran/7813.
- [9] M. Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096.
- [10] H. Yoshida, *On Hermitian forms attached to zeta functions*, in: N. Kurokawa and T. Sunada (eds.), *Zeta Functions in Geometry*, Adv. Stud. Pure Math. **21**, Math. Soc. Japan, Kinokuniya, Tokyo, 1992, 281–325.
- [11] X. Zhu, *Weil positivity in compact windows: a finite reduction, certified two-sided bounds, and a Landau–Widom decay law*, arXiv:2608.24827.
- [12] E. De Jesús, *The Partition Potential and the Laguerre Tower*, Zenodo, 2026. doi:10.5281/zenodo.21108392.
- [13] E. De Jesús, *Riemann Hypothesis: The Weil Enforcer in Audit Coordinates*, Zenodo, 2026. doi:10.5281/zenodo.21115524.